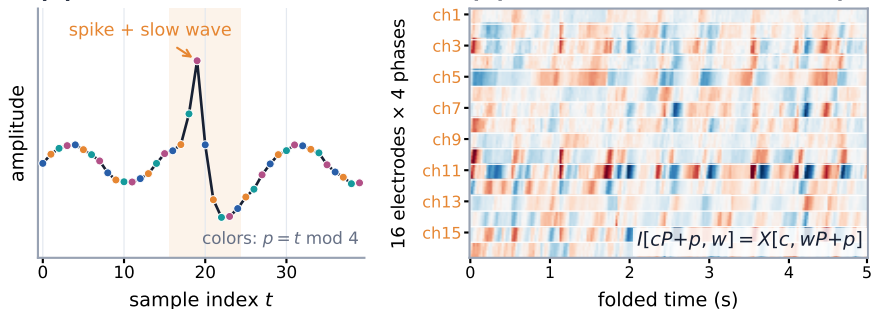
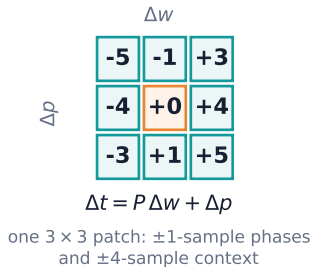


(a) Raw EEG is one-dimensional **(b) Real TUEV: lossless fold ($P = 4$)**



(c) Multi-lag 3×3 filtering



(d) EfficientNet-B0: receptive fields grow with depth

Stem / early

3-7 px $\rightarrow$ 55-140 ms

spikes, slopes,
local phase

Middle

19 px $\rightarrow$ 380 ms

waveform complexes,
short oscillations

Deep

67-147 px $\rightarrow$ 1.3-2.9 s

bursts, rhythms,
cross-electrode

Aggregation

851 px $\rightarrow$ 17 s / token

whole segment
 $\rightarrow$ task decision